SPECIFICATION

Background of the Invention

The performance of conventional computing systems, based on silicon transistors and binary logic, is increasingly limited by fundamental physical constraints known as the "power wall" and the "memory wall." As transistors shrink, the heat generated becomes a primary obstacle to increasing clock speeds. Furthermore, the speed of electrical interconnects between the processor and memory has not kept pace with processing speeds, creating a significant data bottleneck.

The advent of optical interconnects and massively parallel optical buses, as described in related applications, offers a solution to the data transmission bottleneck. These optical buses can carry millions of parallel data channels, encoded in different wavelengths of light, offering bandwidth orders of magnitude greater than electrical buses. However, a new bottleneck has emerged at the point of computation. Current systems must convert this massively parallel optical data back into serial, binary electrical signals for processing by traditional transistor-based CPUs. This conversion step negates many of the advantages of optical transmission, reintroducing latency, heat, and power consumption. The CPU core itself, architected for binary operations, is fundamentally unable to process the rich, high-cardinality data that an optical bus can provide. Therefore, there is a need for a new computing paradigm that can natively process optical, multi-wavelength information.

Summary of the Invention

The present invention provides a system and method for high-radix (or base-N) computing by replacing the binary (base-2) transistor with a wavelength-discriminating phototransistor as the fundamental logic element of a processing unit. The core of the invention is a phototransistor-based logic unit capable of detecting and distinguishing between a plurality of discrete wavelengths of light. Each discrete wavelength is assigned a unique logical state (e.g., Red = 0, Green = 1, Blue = 2, Yellow = 3, etc.), thereby moving beyond the 0/1 limitation of binary systems.

The system comprises a processing unit with arithmetic logic units (ALUs), registers, and control logic built from these wavelength-discriminating phototransistors. This processing unit is integrated on a die and directly coupled to an optical interface, allowing it to receive and transmit data from a parallel optical bus. Because the logic elements can natively understand and process data encoded as multiple colors, the entire optical-to-electrical conversion bottleneck is eliminated. An entire word of high-radix data can be processed in a single clock cycle, leading to an exponential increase in computational density and a dramatic reduction in power consumption and heat generation. This invention provides the foundational processing architecture to complete the end-to-end optical computing ecosystem.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram contrasting a traditional binary transistor logic element with the high-radix phototransistor logic element of the present invention.

FIG. 2 is a system-level block diagram of a high-radix processing unit integrated with an optical bus.

Detailed Description of the Invention

Referring to **FIG. 1**, the fundamental difference between the prior art and the present invention is illustrated. A conventional binary logic element (101) is based on a transistor (102), which can be in one of two states: ON (representing a logical '1') or OFF (representing a logical '0'), based on the presence or absence of an electrical voltage. In contrast, the high-radix logic element of the present invention (103) is based on a wavelength-discriminating phototransistor (104). This phototransistor is designed to receive an input optical signal (105) comprising one of a plurality of discrete wavelengths. The phototransistor (104) can distinguish between these wavelengths and output a corresponding signal indicating a specific logical state (106). For example, in a hypothetical base-4 system, Wavelength 1 could represent state '0', Wavelength 2 state '1', Wavelength 3 state '2', and Wavelength 4 state '3'. This allows for a single logic element to represent four states, doubling the information density compared to a binary element.

Referring to **FIG. 2**, the system architecture of a high-radix processing unit (200) is shown. The processing unit is integrated on a single die. An optical bus (201) provides a massively parallel optical input signal (202), which contains data encoded across a plurality of wavelengths. This signal is received directly by the high-radix processing core (203), which is comprised of the phototransistor-based logic elements described in FIG. 1. The core contains high-radix registers (204) and a high-radix Arithmetic Logic Unit (ALU) (205) capable of performing computations directly on the multi-state data. For example, the ALU can perform an addition operation on two base-4 numbers in a single clock cycle. The processing core (203) can also transmit high-radix optical data (206) back to the optical bus. This direct, native processing of multi-wavelength data eliminates the need for any optical-to-electrical-to-binary conversion, thereby removing a critical performance bottleneck and enabling a more efficient and powerful computing system.